

END TERM EXAMINATION

FIFTH SEMESTER [B.TECH] FEBRUARY 2023

Paper Code: ETCS301

Subject: Algorithms Design and Analysis

Time: 3 Hours

Maximum Marks: 75

Note: Attempt any five questions in all including Q.No.1 which is compulsory.

- Q1 Attempt all questions: (5x5=25)
- a) Define time complexity and space complexity. Write an algorithm for adding n natural numbers and find the space required by that algorithm.
 - b) Define Big 'Oh' notation. Formulate the order of growth. Compare the order of growth $n!$ and $2n$. Differentiate between Best, average and worst case efficiency.
 - c) Differentiate divide and conquer and dynamic programming.
 - d) Explain dynamic programming method of problem solving. What type of problems can be solved by dynamic programming?
 - e) Determine an LCS of $\langle 1, 0, 0, 1, 0, 1, 0, 1 \rangle$ and $\langle 0, 1, 0, 1, 1, 0, 1, 1, 0 \rangle$
- Q2 (4+4+4.5)
- a) Discuss the concepts of asymptotic notations and its properties.
 - b) Analyze the order of growth.
 $F(n) = 2n^2 + 5$ and $g(n) = 7n$. Use the $\Omega(g(n))$ notation.
 - c) Evaluate the recurrence relations.
 - (i). $x(n) = x(n-1) + 5$ for $n > 1$.
 - (ii). $X(n) = x(n/3) + 1$ for $n > 1, x(1) = 1$. (Solve for $n = 3k$)
- Q3 (4+4+4.5)
- a) Which sorting algorithm is best if the list is already sorted? Why?
 - b) Prove that the average running time of Quick Sort is $O(n \log(n))$ where n is the number of elements.
 - c) What are stable algorithms? Which sorting algorithm is stable? Give one example and explain.
- Q4 (4+4+4.5)
- a) Implement UNION using linked list representation of disjoint sets.
 - b) Explain the characteristics of problems that can be solved using dynamic programming.
 - c) Give a control abstraction for Divide and Conquer method. Explain with an example.

P.T.O.

[-2-]

Q5

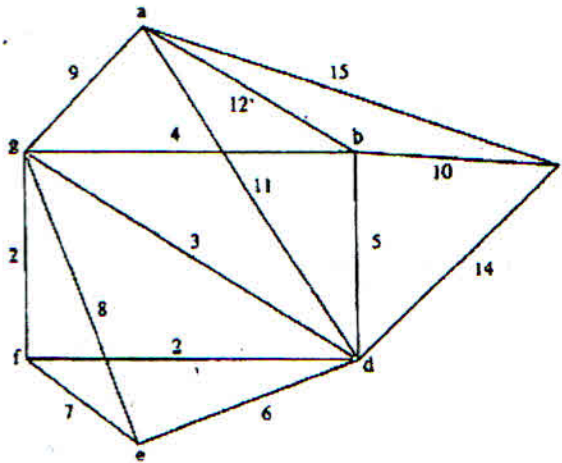
(4+4+4.5)

- Explain the effect of negative weight edges and negative weight cycles on shortest paths.
- Define strongly connected components. How DFS can be used to find strongly connected components?
- Find an optimal paranthesization of a matrix-chain product whose sequence of dimensions is 4×10 , 10×3 , 3×12 , 12×20

Q6

(6+6.5)

- Write Dijkstra's Single Source Shortest path algorithm. Analyze the complexity.
- Find minimum spanning tree for the following graph using Prim's algorithm and discuss complexity.



Q7

(6+6.5)

- Explain Rabin-karp string matching algorithm in brief.
- Find longest common subsequence of following two strings X and Y using any algorithm :
X = 'aabdbacdcb'
Y = 'aabddcbac'

Q8

(4+4+4.5)

- Differentiate between P, NP, NP-completeness and NP - Hard problems.
- How a problem is identified as NP complete problem? Give atleast five problems that can be classified as NP complete problems.
- With examples explain polynomial time reducibility.

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P2/2